

Recitation 12

Image Restoration

Agenda

1. Course Recap (Our Journey)
2. Restoration:
 - Radon
 - Haze
 - Super Resolution
3. Ex3 – The Winners

Course Recap

Video Summarization

סקר הוראה

What did we learn?

- **Foundations**

- How is an image created?
- What can we know about an image, and how can we use it?

- **Time and Frequency Domains**

- What is that?
- 1&2 D Fourier
- Filtering
- STFT
- Derivatives and their properties

What did we learn?

- **Convolution**
 - Useful kernels
 - Fourier and convolutions
- **Image Transformations**
 - Histograms
 - Derivation & edge detection
 - Filtering
 - Fast Forward
 - Basic noise removal

What did we learn?

- **Pyramids**
 - Frequencies
 - Uses
- **Image Denoising**
 - Why?
 - What is success?
 - Bilateral blurring
 - Non local means

What did we learn?

- **Image Warping**
 - Types (translation, rotation etc.)
 - Warping
- **Image Alignment**
 - Direct methods (NCC, LK)
 - Feature points
 - Optical flow
- **Composing Images**
 - Blending
 - Stitching

What did we learn?

- **Hough Transform**
 - How do you represent lines and circles?
 - Using edges, smoothing, gradients etc.
- **Colors**
 - Color subtraction
 - Color spaces
 - Clipping

What did we learn?

- **Geometry**
 - 3D to 2D
 - TTC
 - Epipolar geometry
 - 3D from 2 images, or 1 image
- **Binary Images**
- **NNs**
 - Basics
 - Classifier architectures
 - Generative architectures

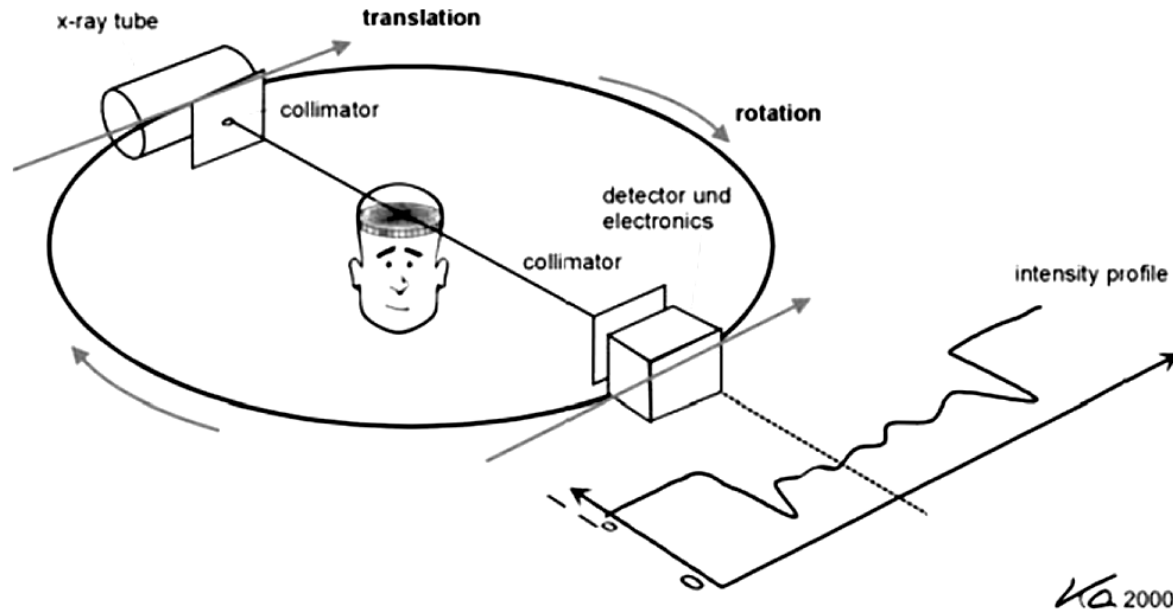
Course Recap

Radon Transform

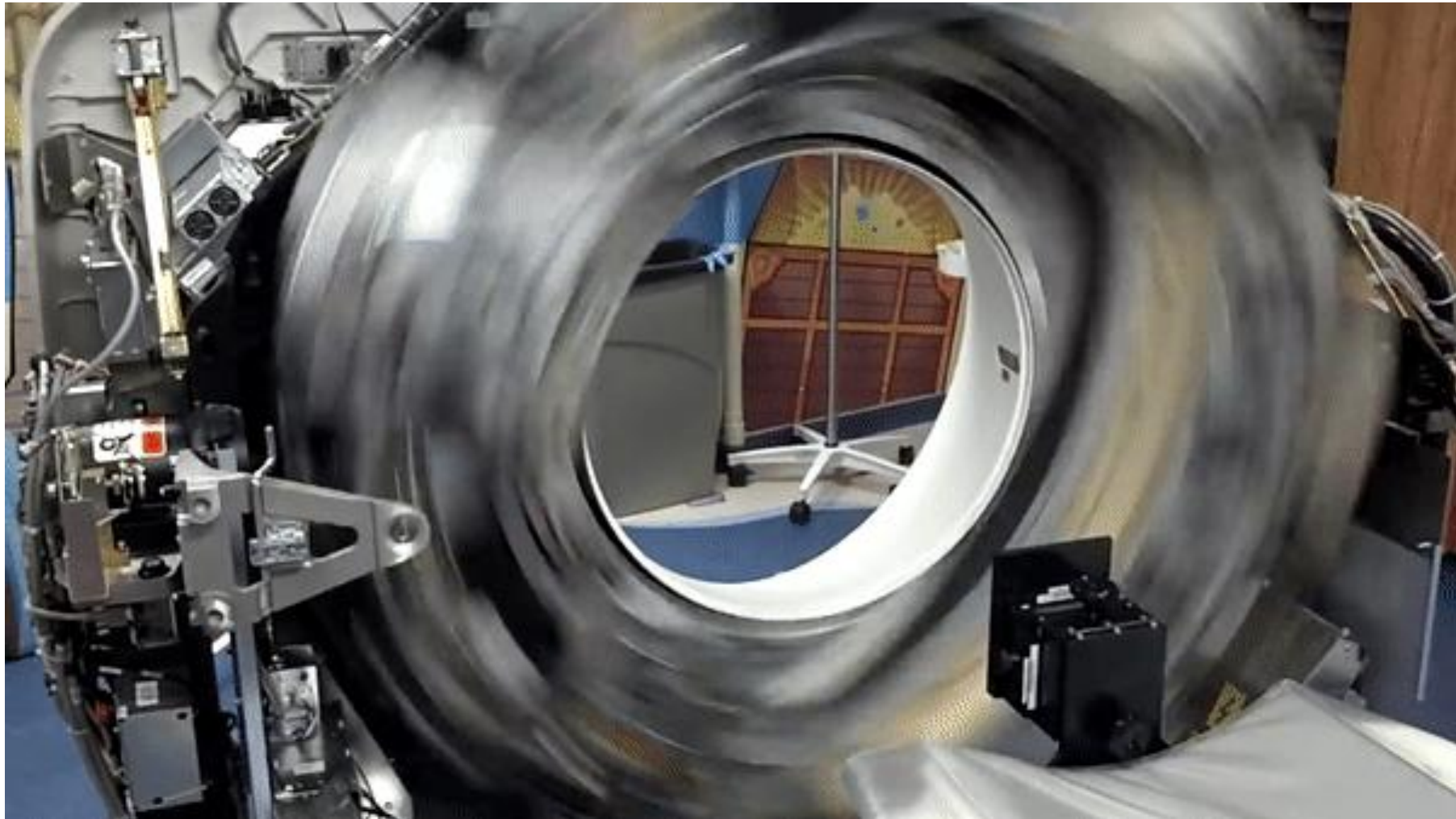
Motivation

The Radon Transform is essential when we aim to understand a multi-dimensional structure (e.g., a 2D or 3D object) but can only acquire its projections from lower-dimensional data (e.g., 1D line integrals).

For example, in medical imaging (CT scans), we wish to reconstruct detailed internal images of the human body using only X-ray projections taken from various angles.

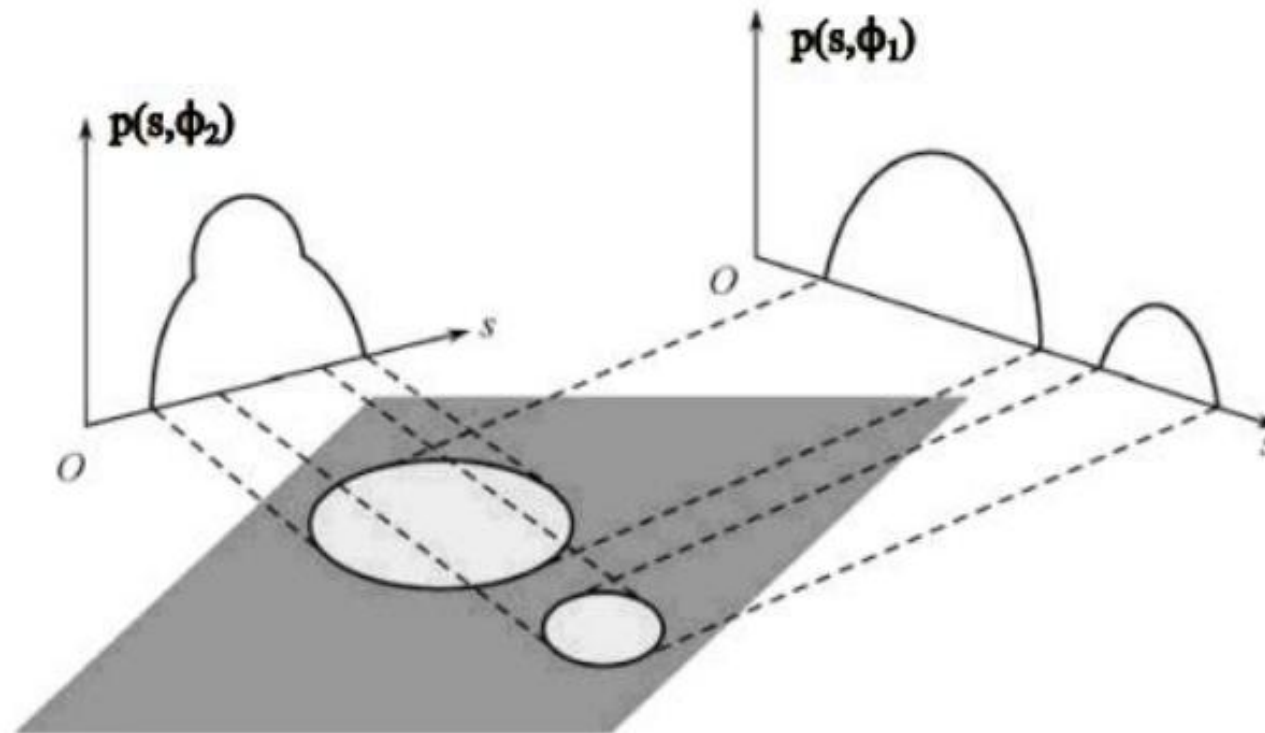


Radon Transform: CT



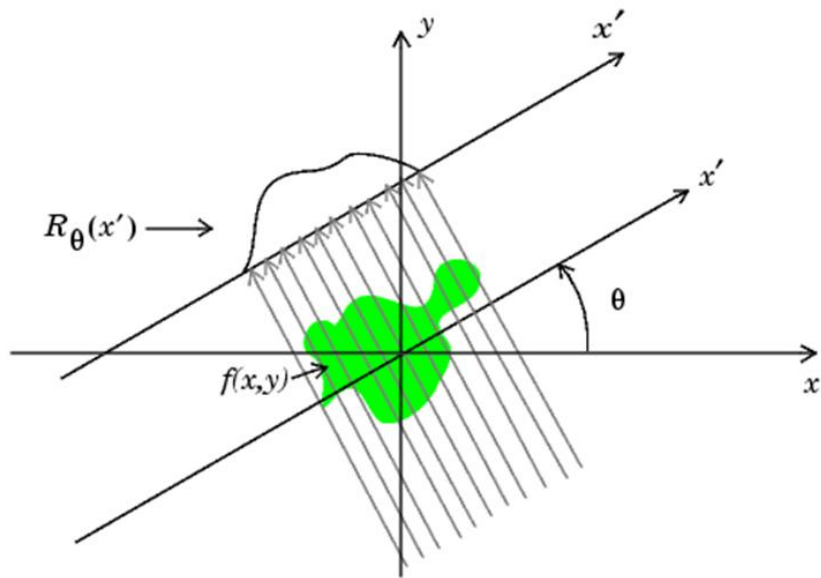
Line Integral Projection

Line integral projections involve integrating an object along lines at different angles, illustrating how the function changes across specific directions.

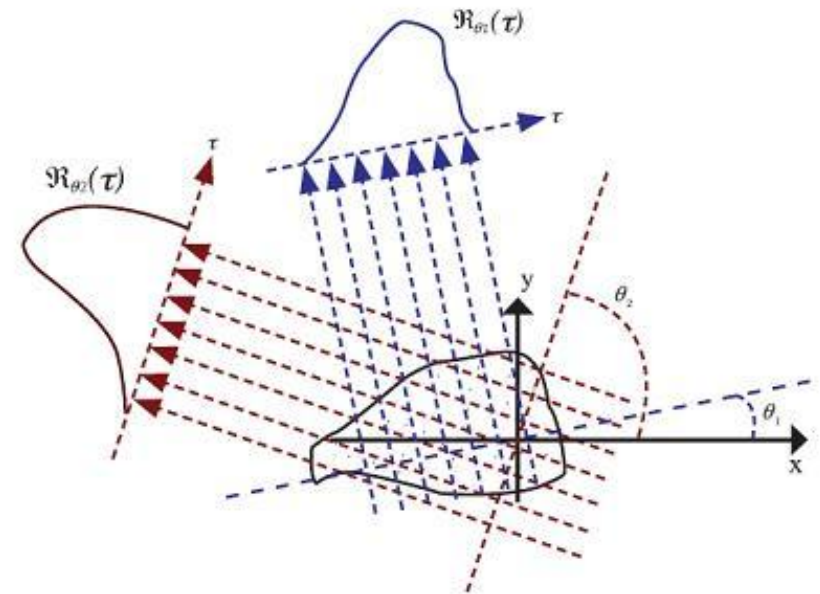


Radon Transform: CT

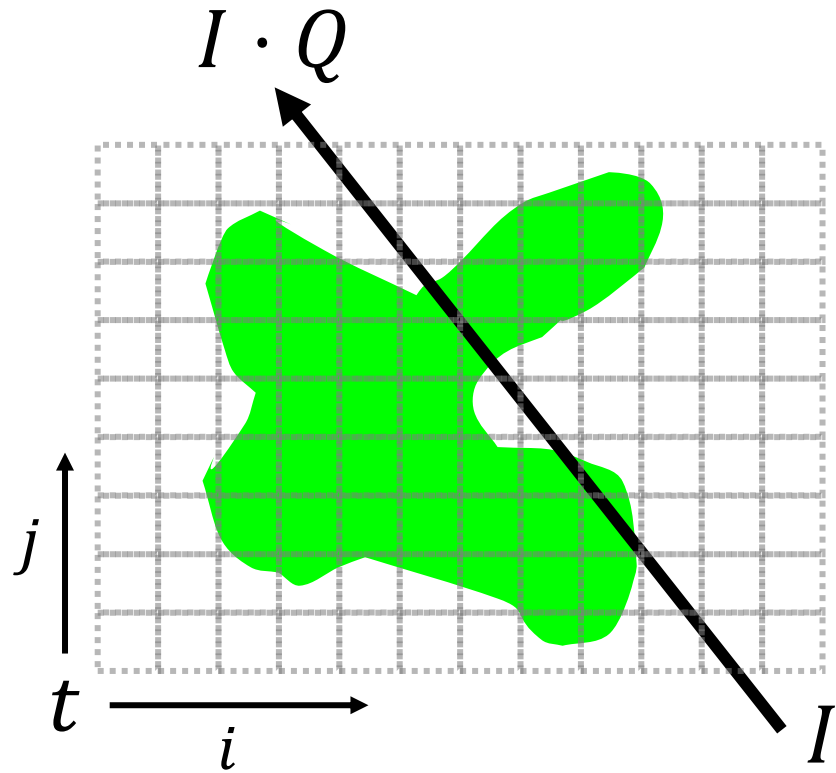
All Projections in One Direction θ



Projections for Multiple Directions



Radon Transform: CT



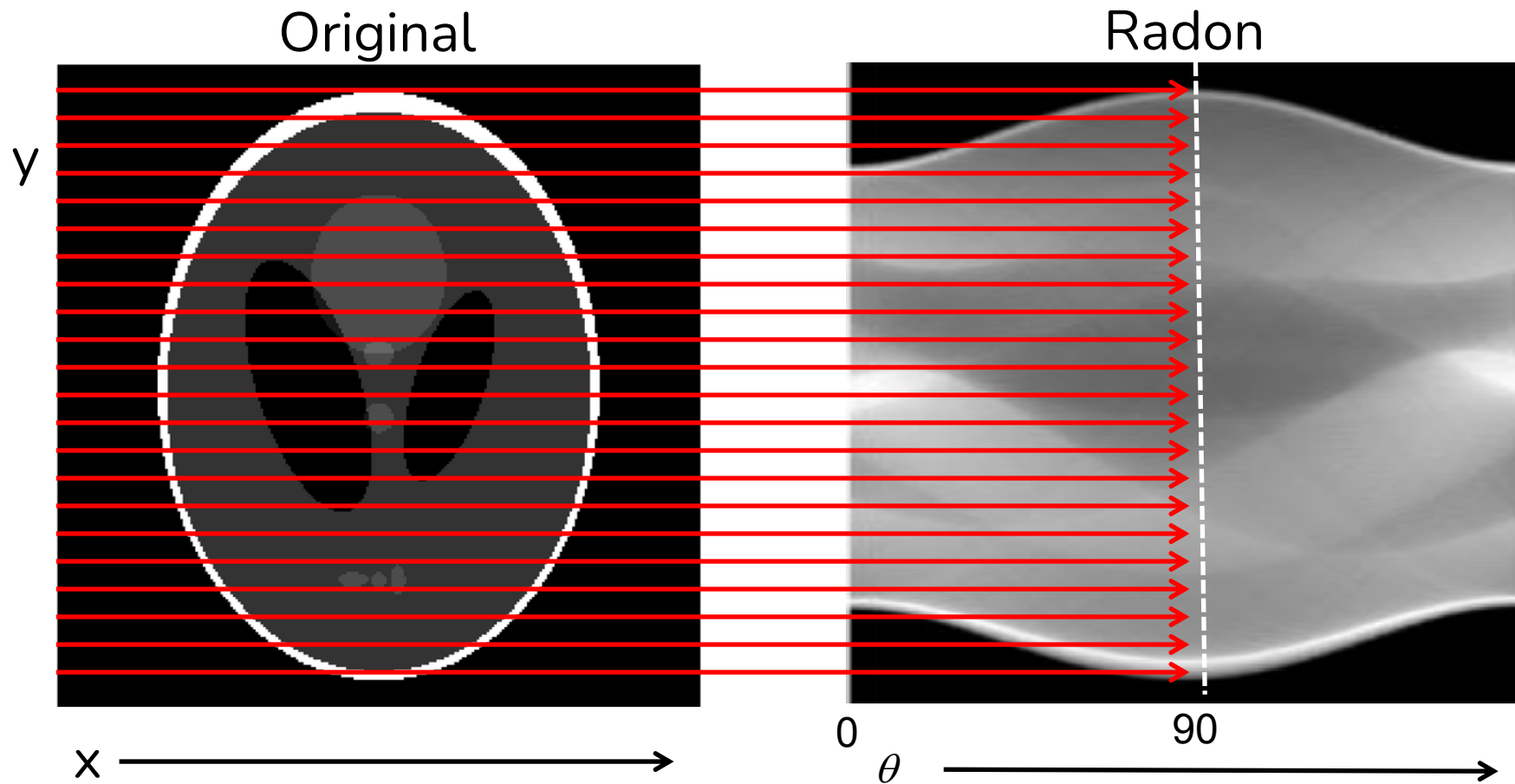
$$Q = \prod_{(i,j) \in \text{line}} t(i,j)$$

$$l(i,j) = \log(t(i,j))$$

$$\log(Q) = \sum_{(i,j) \in \text{line}} l(i,j)$$

Radon Transform: CT

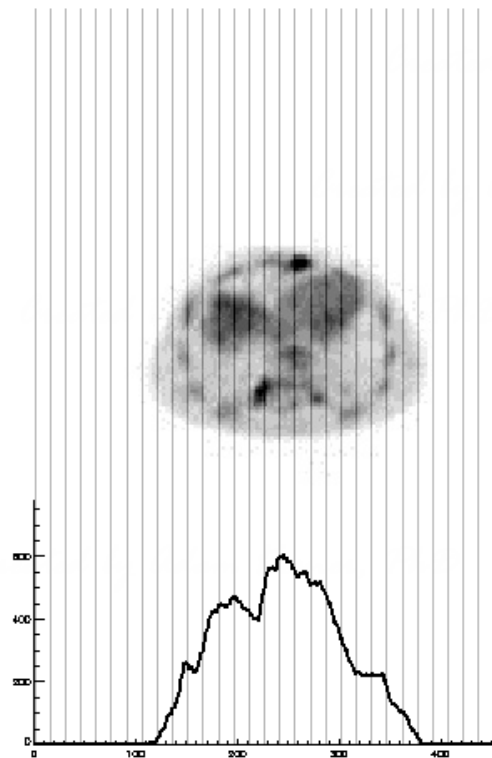
Full Radon Transform



Radon Transform: CT

(True) Emission Volume

Sinogram (stored data)



Forward
Projection

angle
0

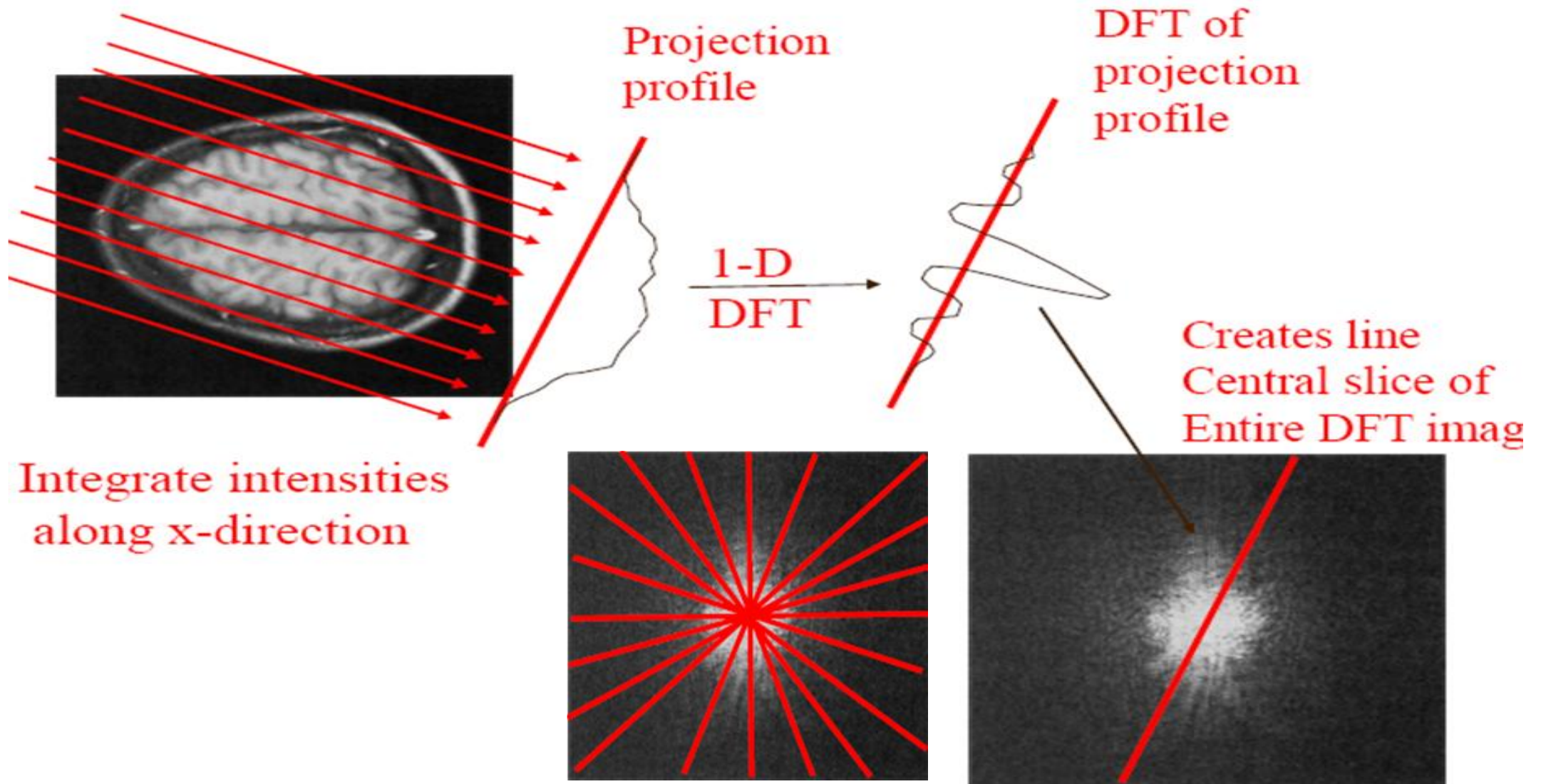
Theta (angle)

Rho (offset)

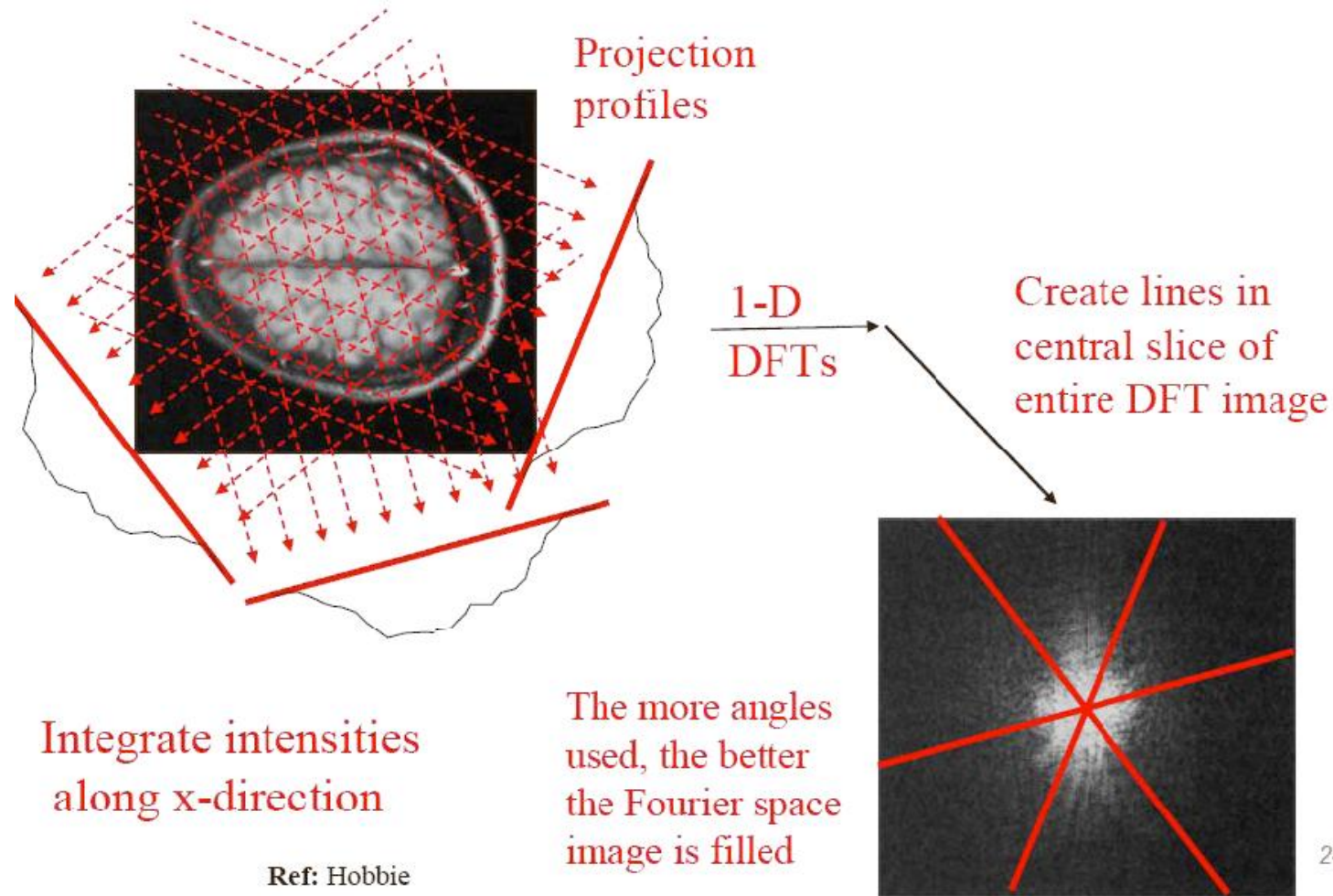
intensity profile:



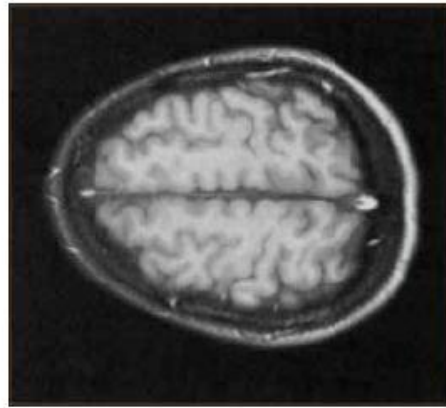
Fourier Slice Theorem - One Slice



Fourier Slice Theorem - Multiple Slices

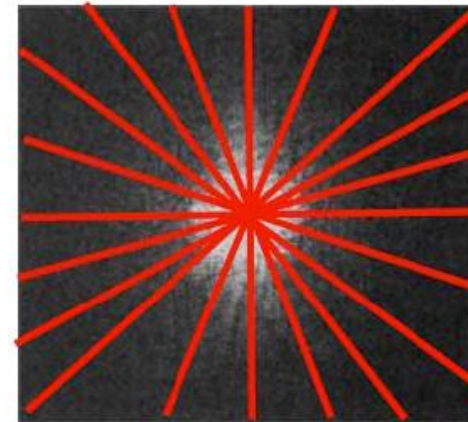


Inverse Fourier from Collected Slices



DFT image represents integration of original projections DFT transformed and summed together.

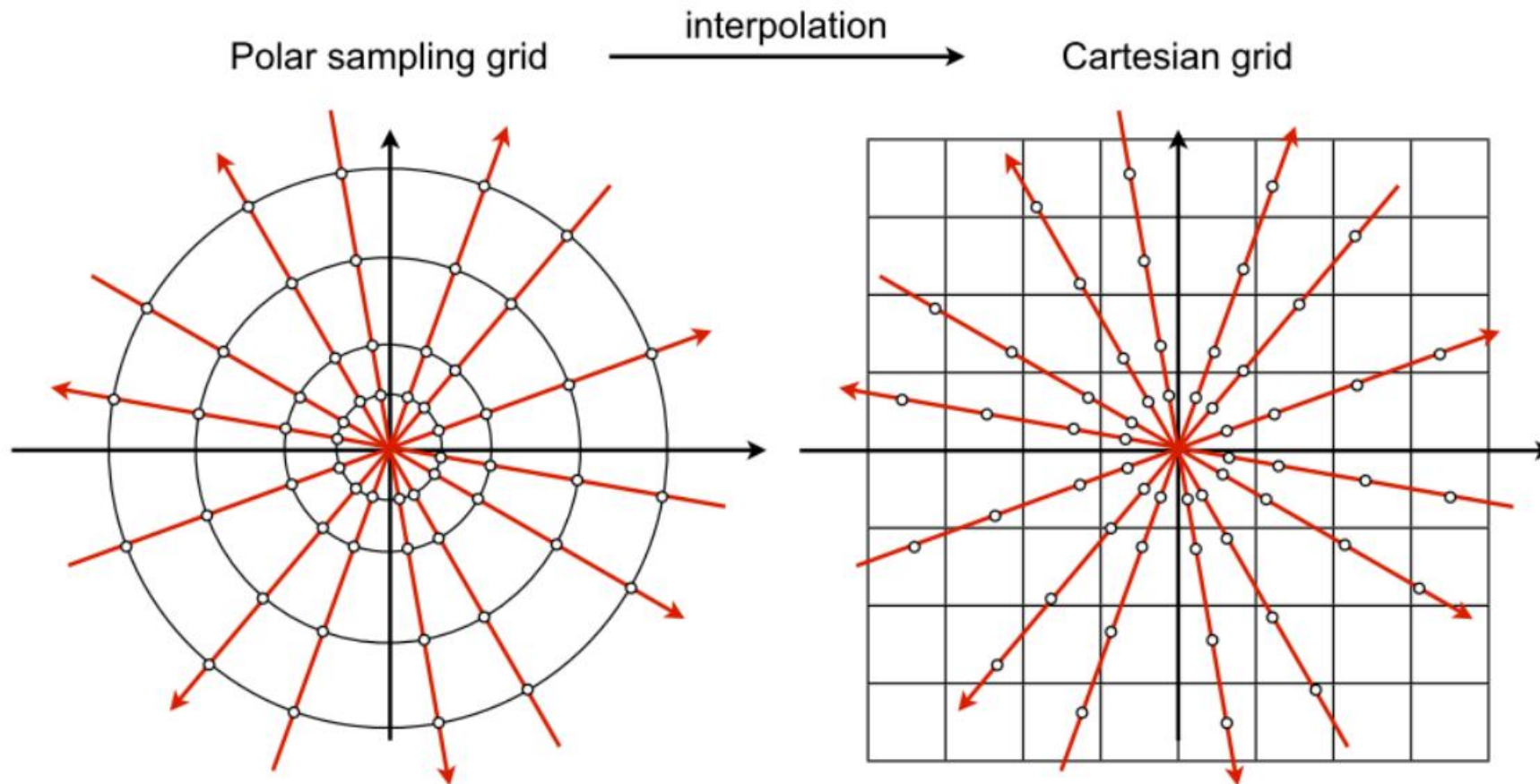
1-D
DFTs
at each
projection



This is the fast way to create the DFT image from projection data. The more projections taken, the more complete the sampling.

Ref: Hobbie

Problem



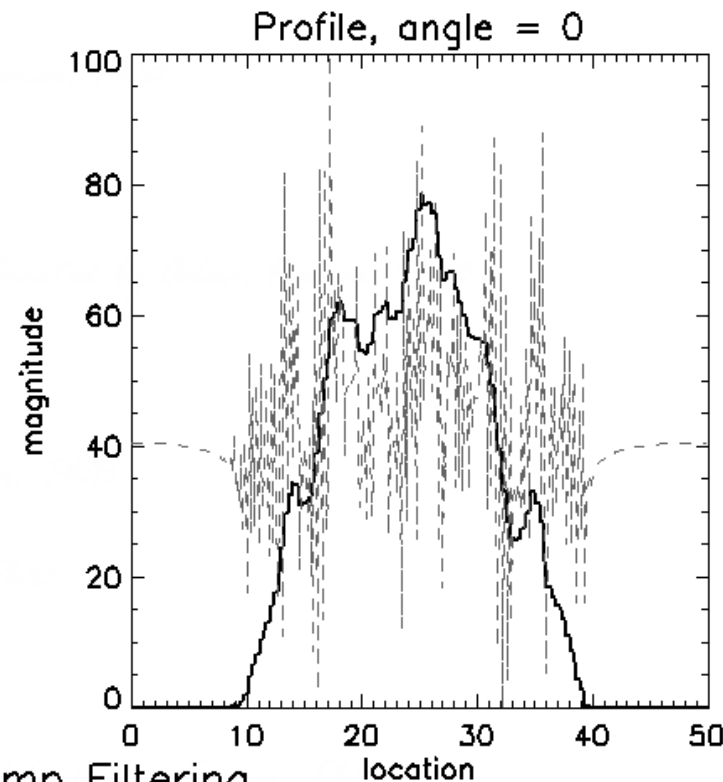
Filtered Back Projection (FBP)

Step 1 – Filter (Ramp Filter)

$$H(w) = |w|$$

$$\hat{F}(w) = H(w)F(w) = |w|F(w)$$

Sinogram



Sinogram Ramp Filtering

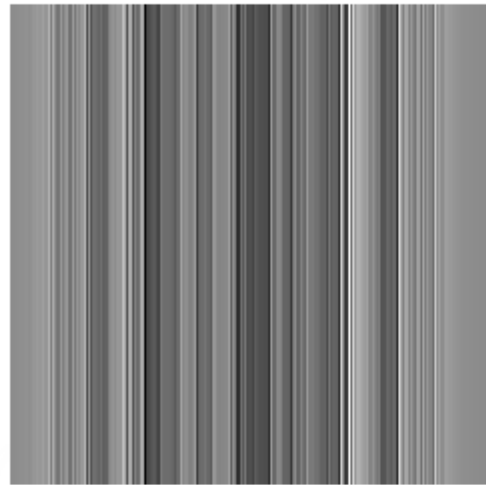
From:

<https://humanhealth.iaea.org/HHW/MedicalPhysics/NuclearMedicine/ImageAnalysis/3Dimagereconstruction/index.html>

Filtered Back Projection (FBP)

Step 2 – Back Projection

Reconstructed image



Filtered Sinogram



←
Filtered Back Projection

From:
<https://humanhealth.iaea.org/HHW/MedicalPhysics/NuclearMedicine/ImageAnalysis/3Dimagereconstruction/index.html>

Filtered Back Projection (FBP)

Reconstructed image



Percent
backprojected
5 %
from selected
angles

Sinogram



Theta (angle)

Rho (offset)

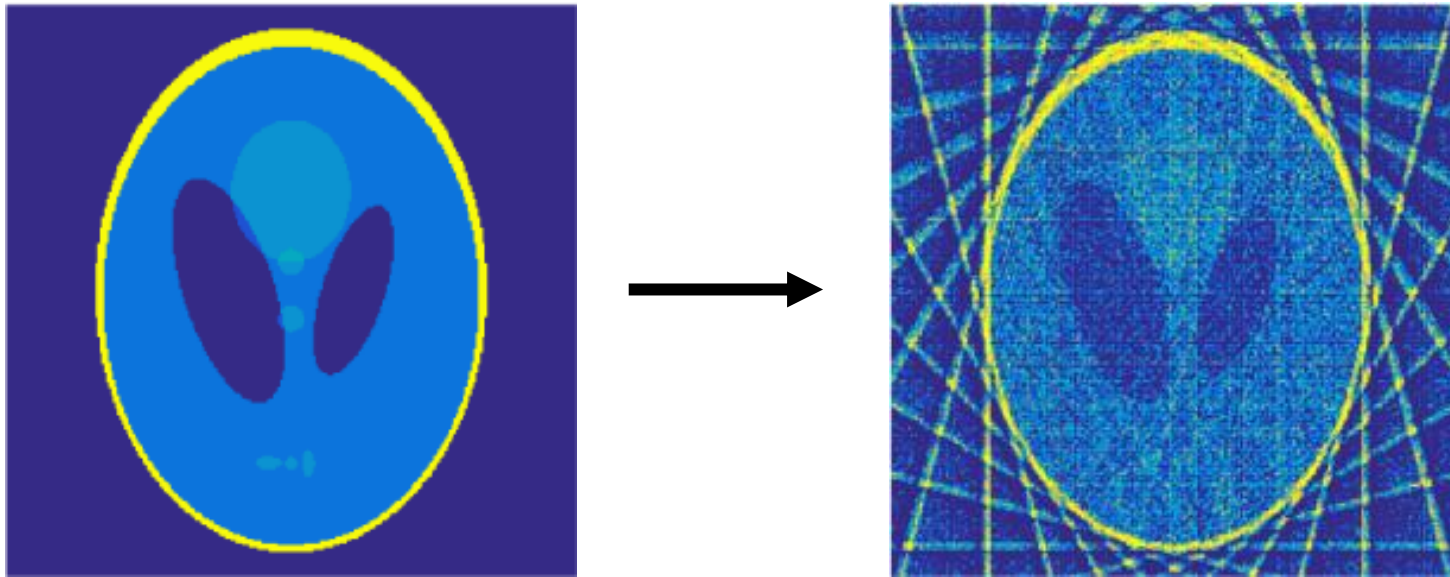
←
Filtered Back Projection

From:

<https://humanhealth.iaea.org/HHW/MedicalPhysics/NuclearMedicine/ImageAnalysis/3Dimagereconstruction/index.html>

Filtered Back Projection (FBP)

If the angles between each sample **are too large**
e.g., $0, 15^\circ, 30^\circ, \dots, 180^\circ$ degrees, etc.



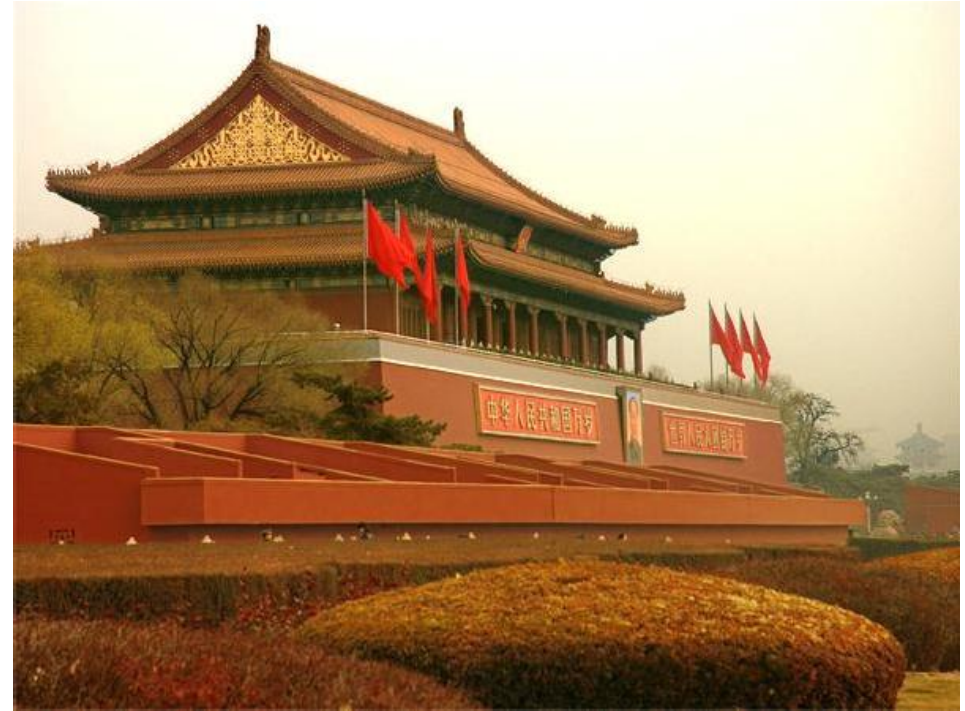
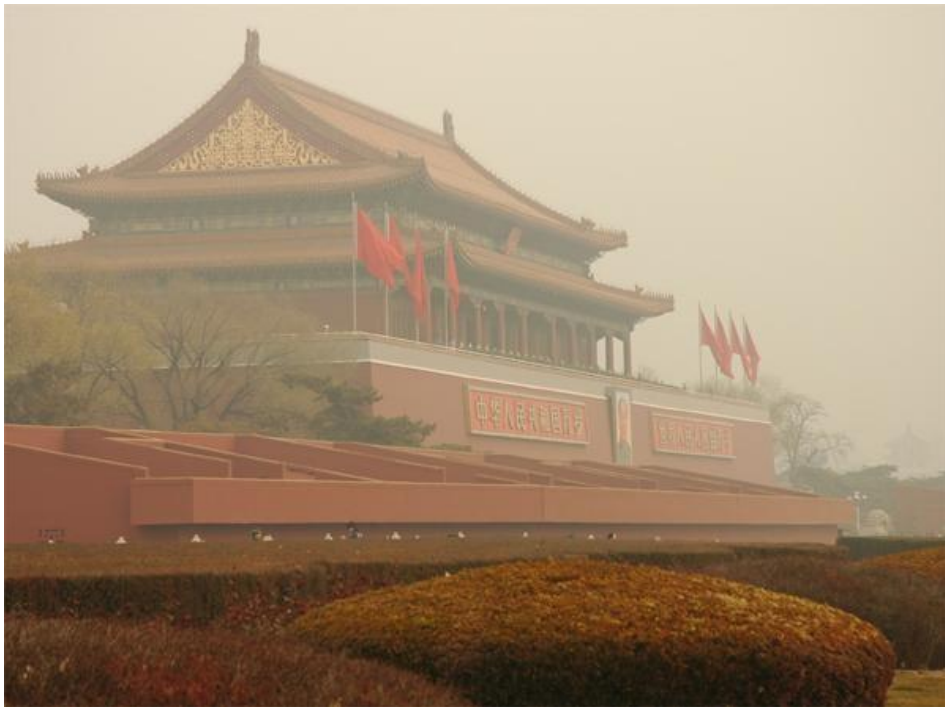
Course Recap

Haze Removal

Haze Removal

Motivation:

We aim to eliminate haze from images.



Priors in Computer Vision

In computer vision, we often rely on prior knowledge about the world, referred to as a "prior".

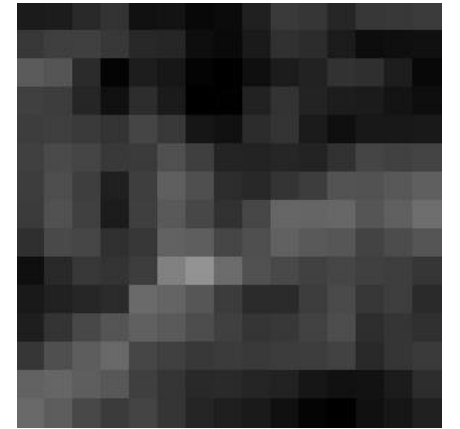
For this problem, we will utilize the **Dark Channel Prior**.

Dark Channel

$$\min(RGB, local\ patch)$$

$$J^{dark}(x) = \min_{y \in W(x)} \left(\min_{c \in \{R, G, B\}} J^c(y) \right)$$

W – 15X15 Window in our case



Original Image



$\min(R, G, B)$



Dark Channel



A Surprising Observation

Haze-free



A Surprising Observation

Haze-free



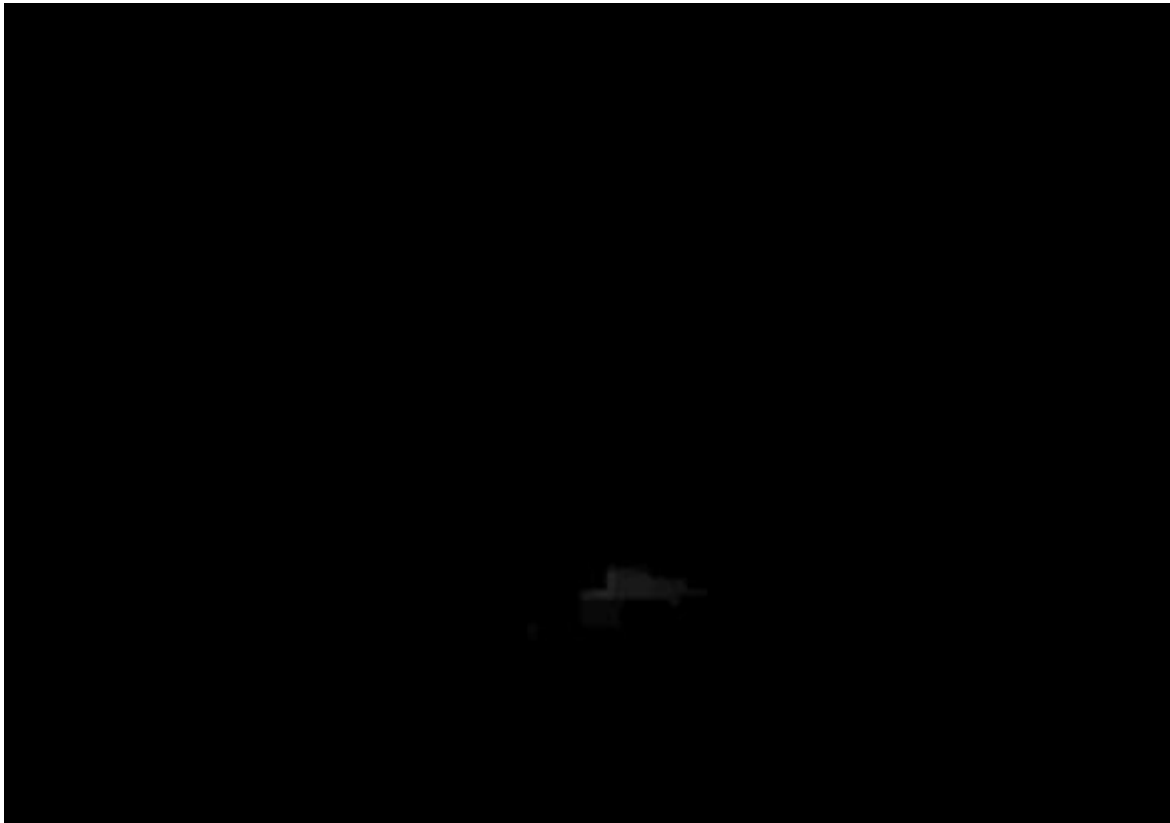
A Surprising Observation

Haze-free

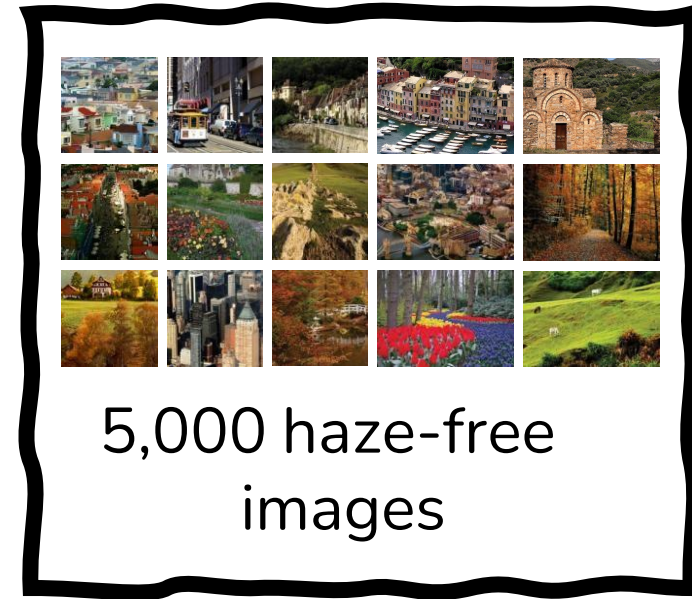
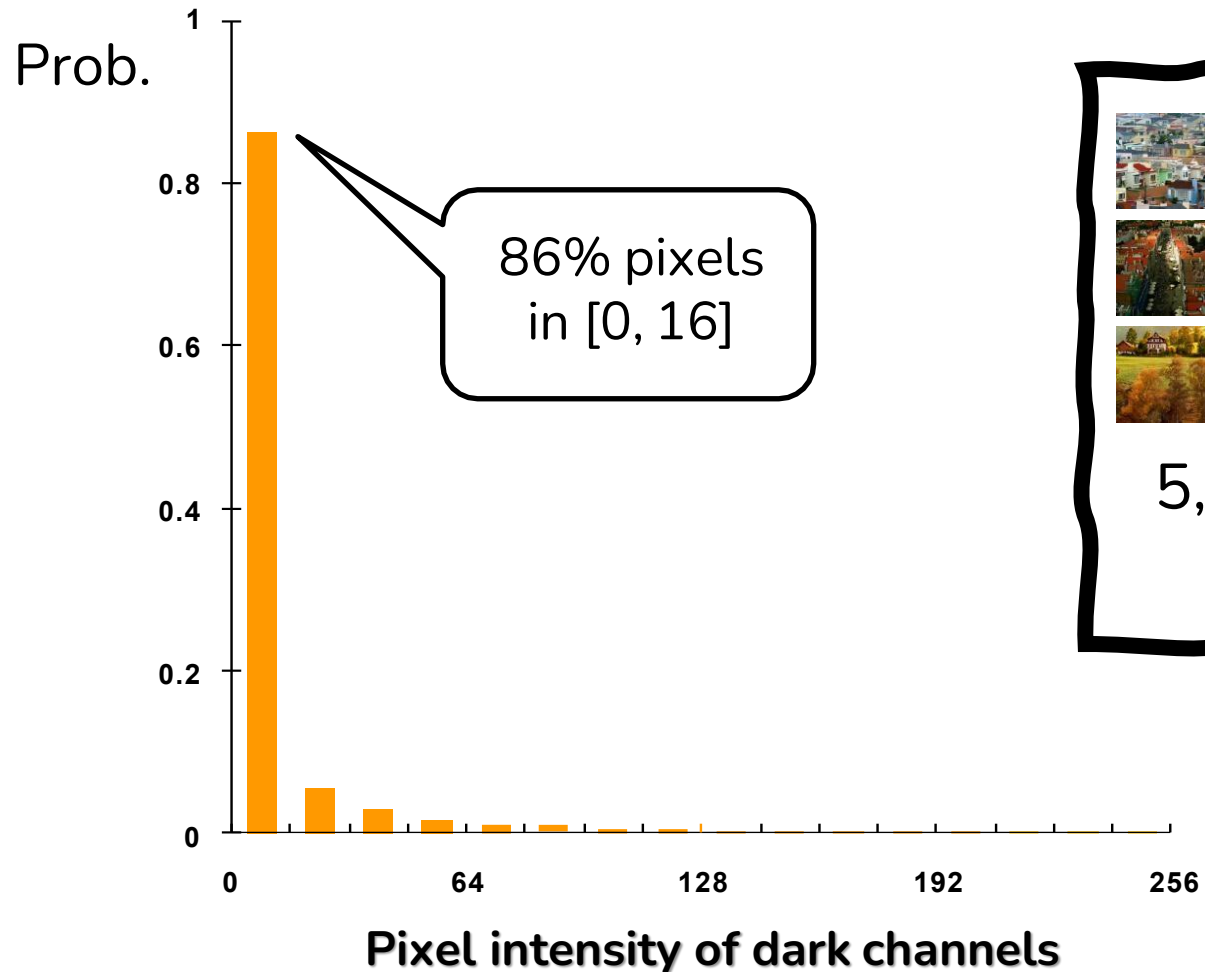


A Surprising Observation

Haze-free



A Surprising Observation



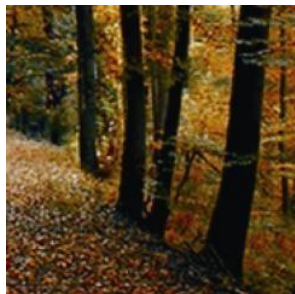
What makes it dark?



Shadow



Colorful Objects



Black Objects

Dark Channel of Hazy Image



Hazy image



Dark Channel

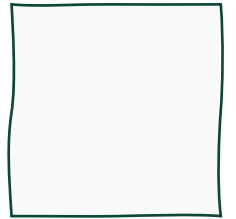
**The dark channel
Is no longer dark.**

Scene Radiance Restoration

We will assume that the atmosphere is either white or a shade of gray that is very close to white.

$$I = J \cdot t + A \cdot (1 - t)$$

Atmospheric light



Hazy image



Scene without haze



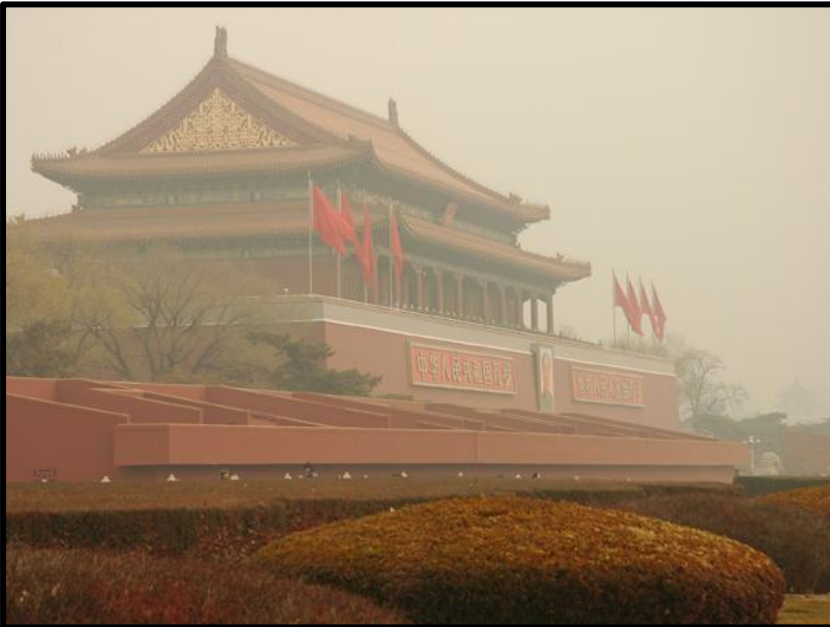
Transmission (t)



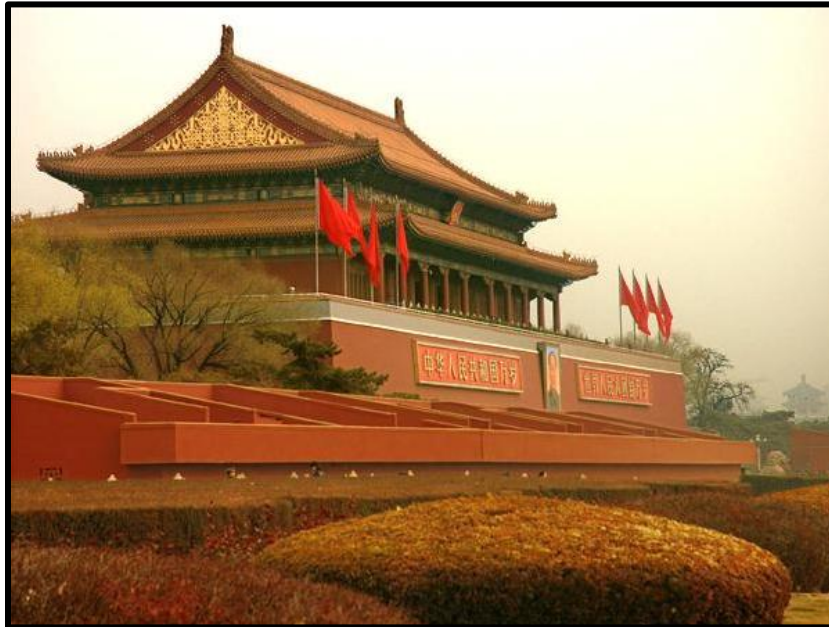
Homogenous Atmosphere

$$t(x) = e^{-\beta d(x)}$$

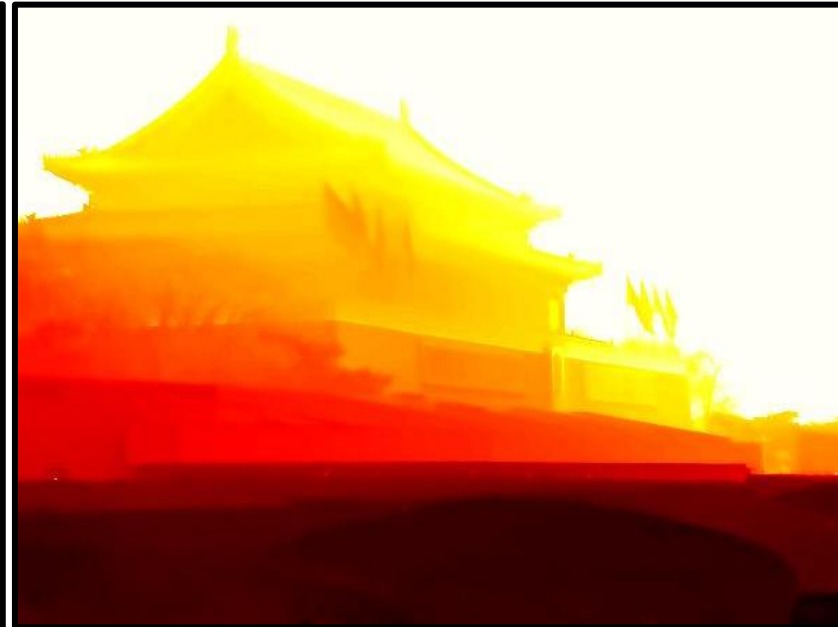
Hazy image



Without haze



Depth



More Results



More Results



More Results



More Results



More Results



More Results



Course Recap

Multi-Frame Super Resolution

Basic Multi-Frame Super-Resolution Idea

Given: A set of low-quality images, where each image is offset by a fraction of a pixel relative to the others



Required: Fusion of these images into a higher resolution image

Basic Multi-Frame Super-Resolution Idea

Given: A set of low-quality images, where each image is offset by a fraction of a pixel relative to the others



Required: Fusion of these images into a higher resolution image



Comment: This is an actual super-resolution reconstruction result. Does it really give more resolution?

Example - Enhanced Mosaics

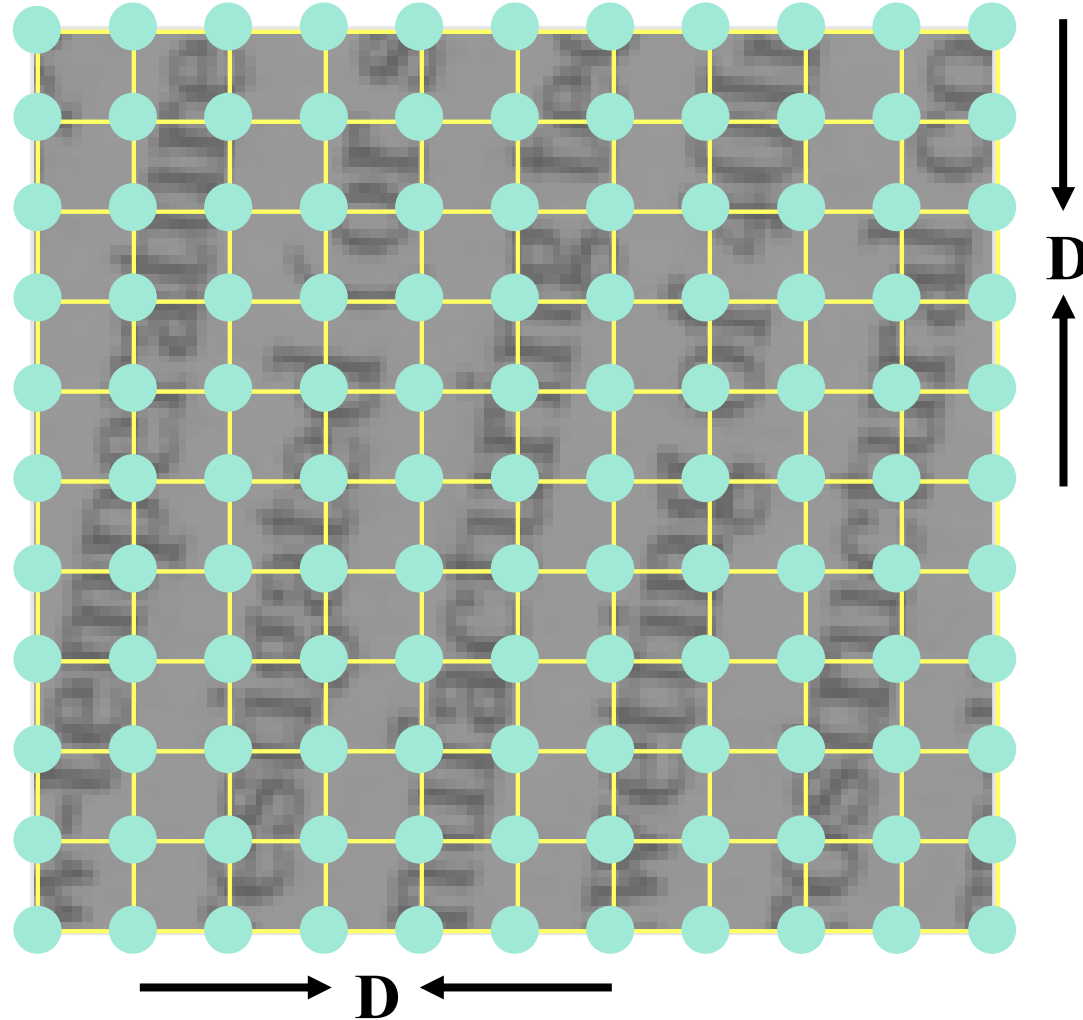


Example - Enhanced Mosaics



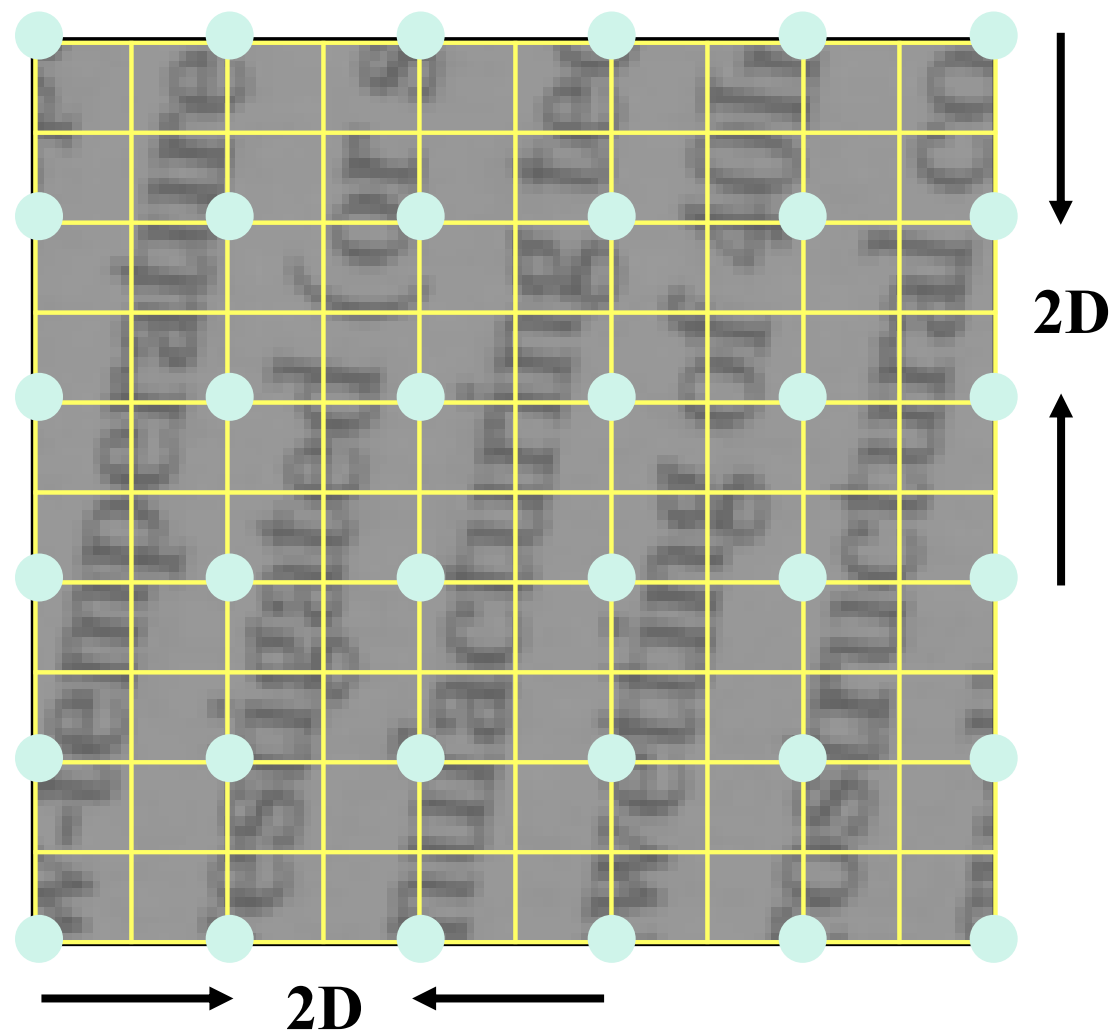
Intuition

We would like to get a uniform sampling with sampling distance $\leq D$



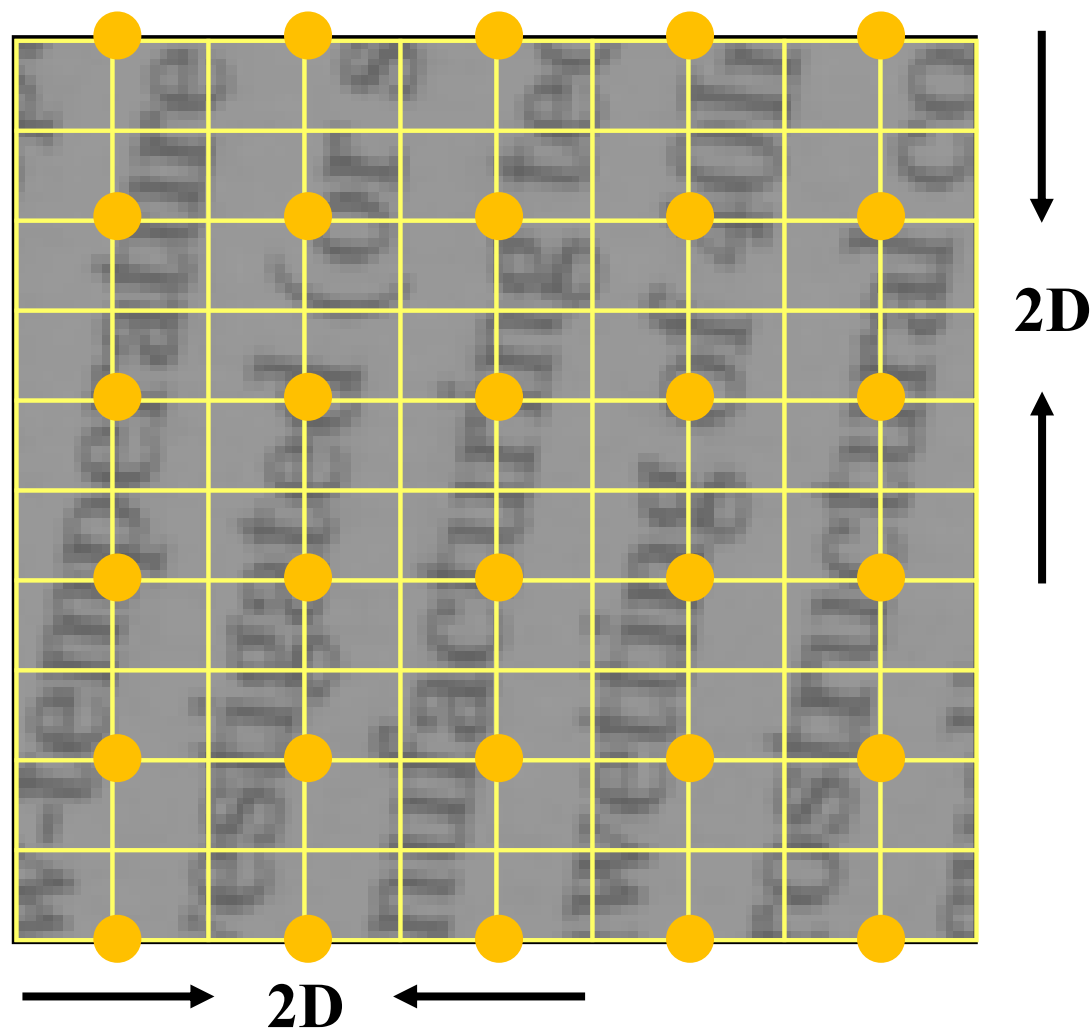
Intuition

Due to our limited camera resolution, we sample using an insufficient 2D grid



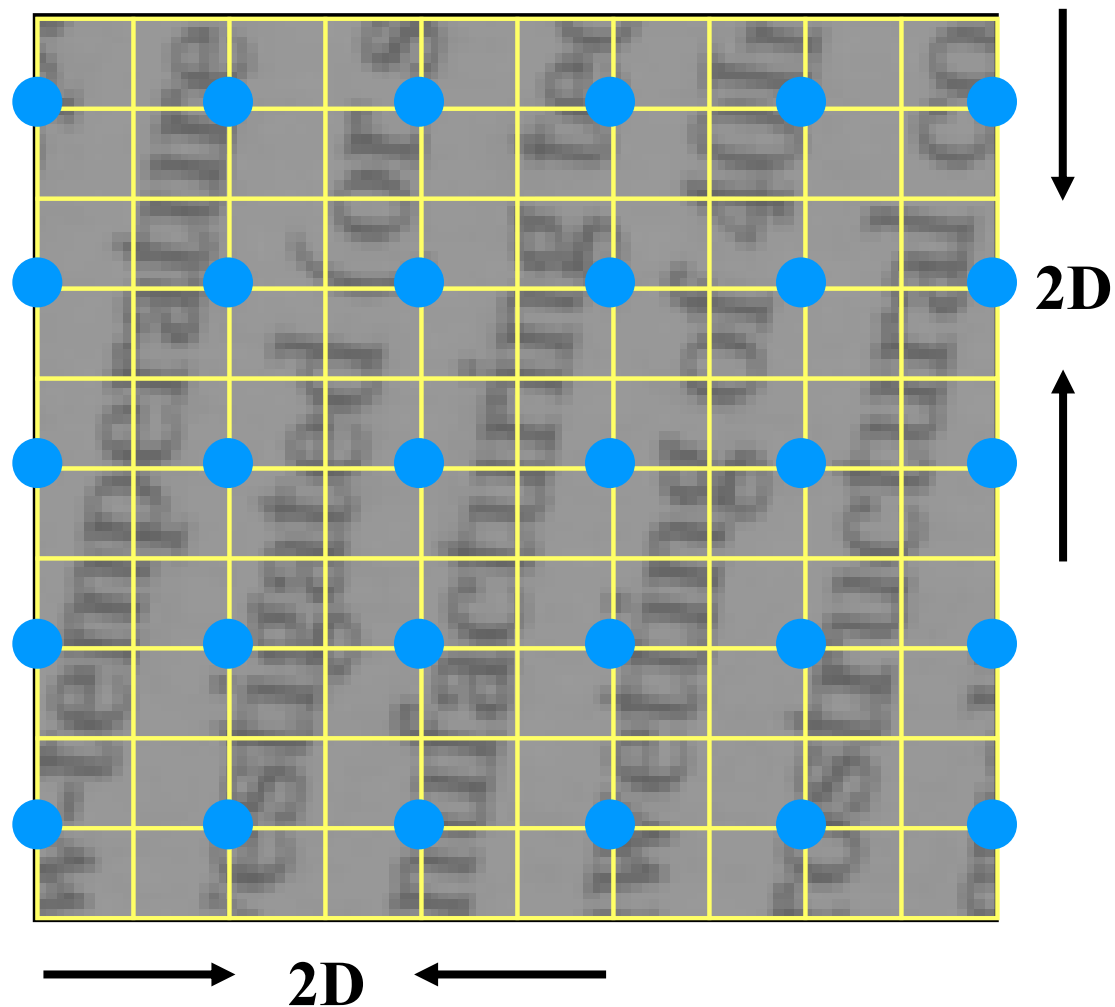
Intuition

However, if we take a second picture, shifting the camera 'slightly to the right' we obtain:



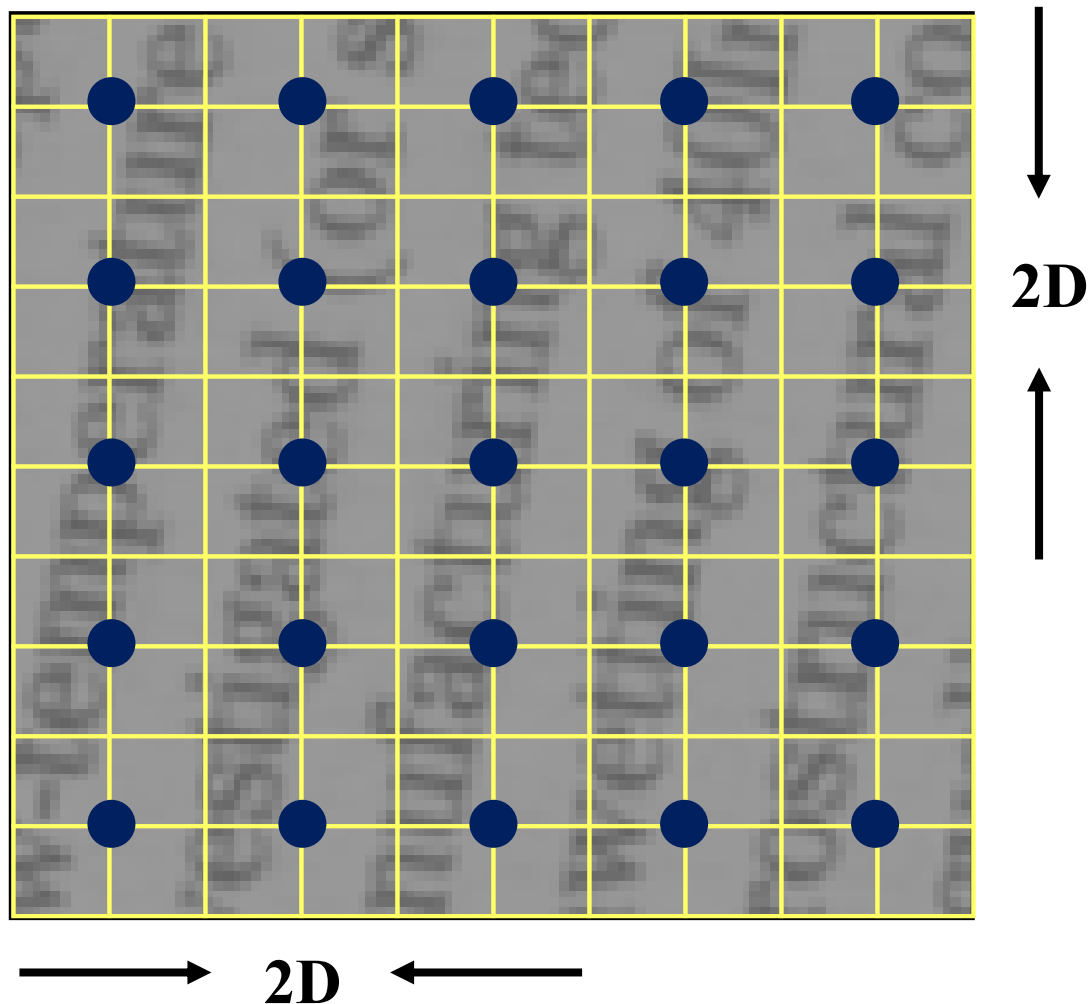
Intuition

Similarly, by shifting down we get a third image:



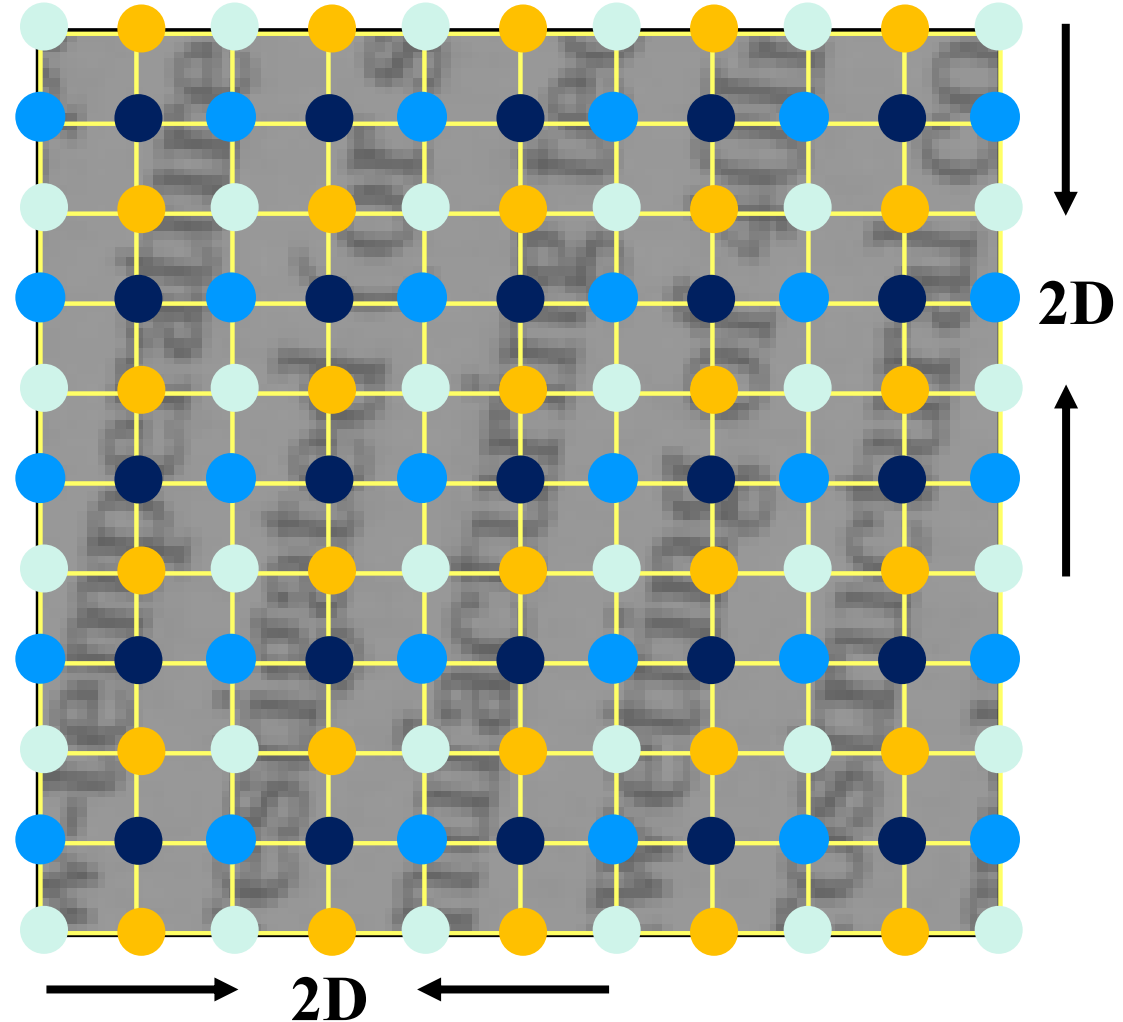
Intuition

And finally, by shifting down and to the right we get the fourth image:

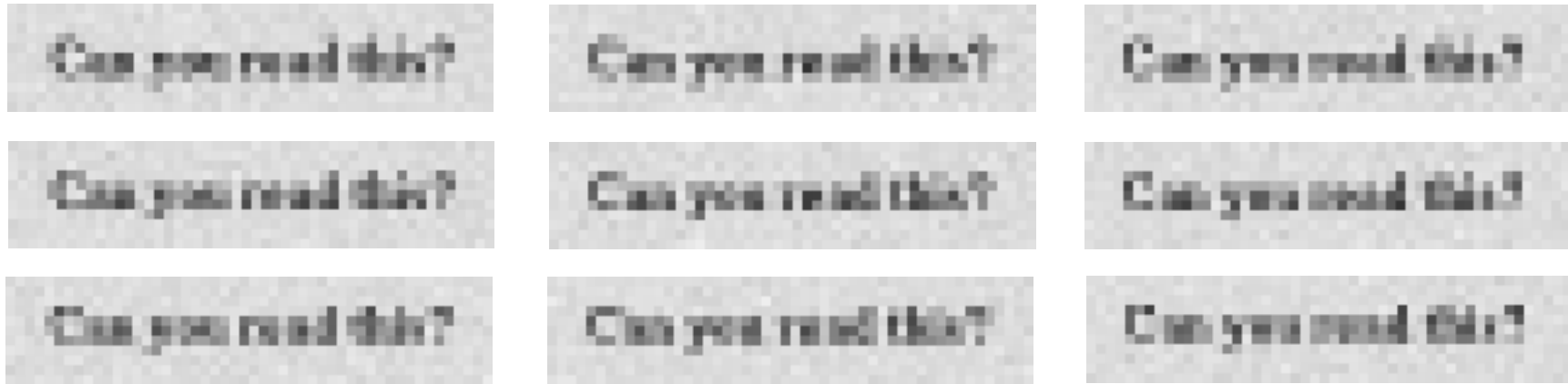


Intuition

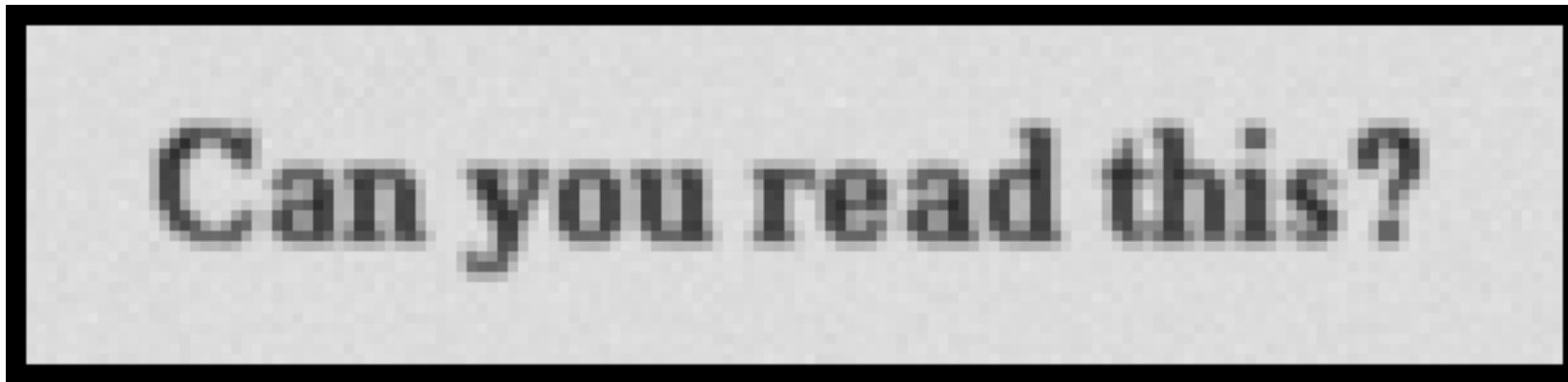
It is trivial to see that interlacing the four images, we get that the desired resolution is obtained, and thus perfect reconstruction is guaranteed.



A Small Example

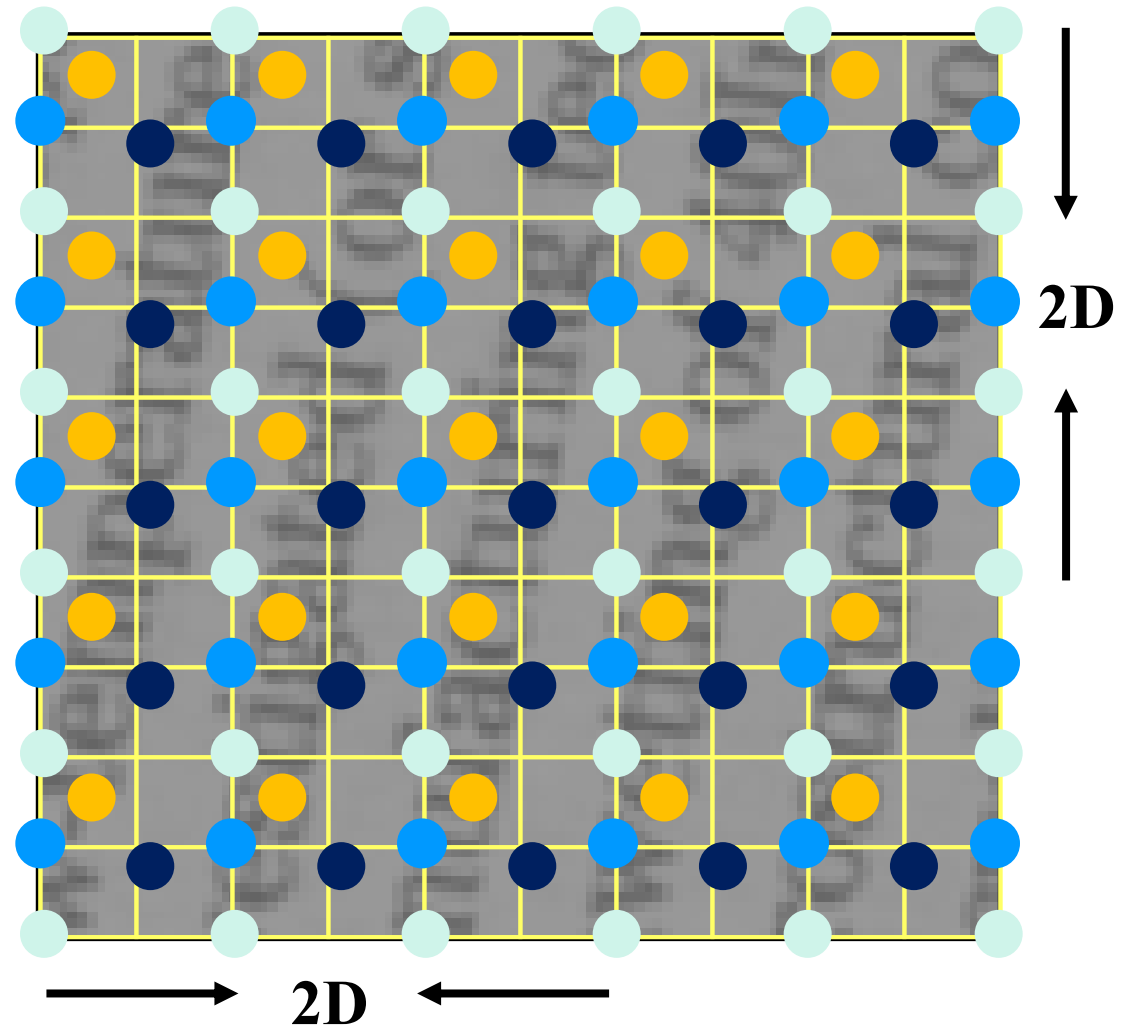


3:1 scale-up in each axis using 9 images
Translation of $\frac{1}{3}$ pixel between images
Uniform blur during imaging



Problem

No one guarantees that the offsets will be convenient (such as exactly half or one-third of a pixel).



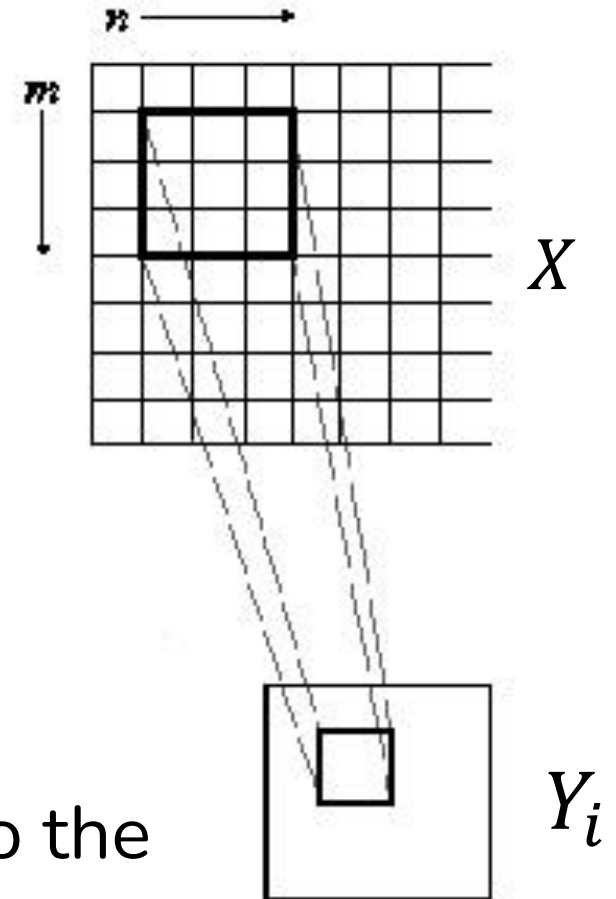
Using Error Propagation

Why? We can assume the observed low-resolution images Y_i are generated from the high-resolution image X by translating, blurring, and sampling.

Given the LR images Y_i , we look for the HR image X whose simulated projections $P_i(X)$ are closest to Y_i

$$E(X) = \sum_{i=1}^N \|P_i(X) - Y_i\|^2$$

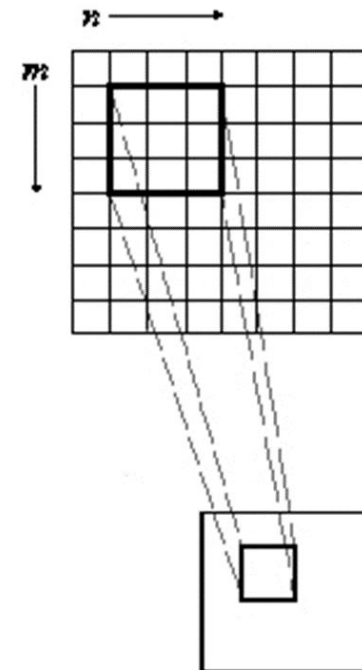
P_i - is the simulated known projection of image X onto the grid of the image Y_i . P_i is computed by motion analysis.



Using Error Propagation

Steps:

1. Calculate sub-pixel motion between the LR images Y_i
2. Initial guess for the HR image X , for example, can be the average of all aligned and enlarged LR images Y_i
3. **Back Projection process**, Repeat the following for all Y_i :
 - For each LR image Y_i , compute projection error $P_i(X) - Y_i$
 - Change pixel values of image X to minimize above error by back projecting the error.



Single Frame Super resolution

Steps:

1. Given a single LR image Y , create the “best” HR image X
2. Information is taken from
 1. Other images
 2. Other parts of same image
3. Best practice is to train a NN to “upscale” images by giving it pairs of synthetically generated LR-HR images, or blurred-sharp images.
4. Results are “hallucinated”, and do not necessarily reflect the true scene, as the details were taken from other places.

Good Luck!